The Matter With Things: Our Brains, Our Delusions and the Unmaking of the World

Iain McGilchrist

142

the brain's left hemisphere is designed to help us ap-prehend – and thus manipulate – the world; the right hemisphere to com-prehend it – see it all for what it is.

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as the great physicist Erwin Schrödinger put it in Science & Humanism, it seems plain and self-evident, yet it needs to be said: the isolated knowledge obtained by a group of specialists in a narrow field has in itself no value whatsoever, but only in its synthesis with all the rest of knowledge and only inasmuch as it really contributes in this synthesis toward answering the demand, τίνες δὲ ἡμεῖς; 'Who are we?' 3

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At the core of the contemporary world is the reductionist view that we are – nature is – the earth is – 'nothing but' a bundle of senseless particles, pointlessly, helplessly, mindlessly, colliding in a predictable fashion, whose existence is purely material, and whose only value is utility. Neither Plotinus nor Schrödinger would have been impressed.

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Reductionism can mean a number of things, but here I mean quite simply the outlook that assumes that the only way to understand the nature of anything we experience is by looking at the parts of which it appears to be made, and building up from there. By contrast I believe that the whole is never the same as the sum of its 'parts', and that, except in the case of machines, there are in fact no 'parts' as such, but that they are an artefact of a certain way of looking at the world.

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Reductionism envisages a universe of things – and simply material things at that. How these things are related is viewed as a secondary matter. However, I suggest that relationships are primary, more foundational than the things related: that the relationships don't just 'connect' pre-existing things, but modify what we mean by the 'things', which in turn modify everything else they are in relationship with.

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complexity is the norm, and simplicity represents a special case of complexity, achieved by cleaving off and disregarding almost all of the vast reality that surrounds whatever it is we are for the moment modelling as simple (simplicity is a feature of our model, not of the reality that is modelled).

286

Some scientists, whether they put it this way or not when they are asked to reflect, still carry on as if there just exists a Reality Out There (ROT), the nature of which is independent of any consciousness of it: naïve realism. These are usually biologists; you won't find many physicists who would think that. In reality, we participate in the knowing: there is no 'view from nowhere'.

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Meanwhile, on the other hand, there are philosophers of the humanities who think that there is no such thing as reality, since it's all Made Up Miraculously By Ourselves (MUMBO): naïve idealism.

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The idea of a Gestalt is central to this book: by it I mean the form of a whole that cannot be reduced to parts without the loss of something essential to its nature. Indeed, what I hope to offer in this book is just such a Gestalt – one that is based on an understanding of the import of the structure of our brains.

353

The flow of the universe is always creative, though it has order, and is not random or chaotic; the world is always a matter of responsiveness, though it is equally not a free-for-all. It is a process of creative collaboration, of co-creation.

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Music happens to be a very clear case of how what we take to be a thing emerges from a complex of relationships, both those between notes and those between individual consciousnesses. But all our experience, not just in music but in life, both mental and physical , is of such a complex flow, a constantly unfolding, responsive dance of reciprocal gestures.

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To the left hemisphere, you find the truth about something by building it up from bits. But, as the right hemisphere is aware, to understand it you need to experience it as a whole, since the whole reveals as much about the nature of the parts as the parts do about the nature of the whole.

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The choice we make of how we dispose our consciousness is the ultimate creative act: it renders the world what it is. It is, therefore, a moral act: it has consequences. 'Love', said the French philosopher Louis Lavelle, 'is a pure attention to the existence of the other'.15

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the nature of the difference between the hemispheres is far from anything hitherto imagined, and that it fulfils an evolutionary purpose of the utmost importance. It is not a separation of reason from emotion, or language from visuo-spatial skills, or any of the other things that used to be said, since, indeed, each hemisphere deals with absolutely everything – just in a reliably different way.

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The type of attention, then, is critical to what it is that one sees when one encounters the world. And encountering the brain is no exception. If one attends to it in one way, one sees a piece of machinery. If one attends in another, one sees part of a person. Crucial difference, for reasons I have already touched on. Ask the 'machine' question, and you find that each hemisphere is involved with 'processing' language, reason, music, emotion, imagery, morality, the self and much else besides. This could lead, and has led, people to say that there is no difference between the hemispheres. But each is involved with all these areas in a consistent, and consistently very different, way, as I demonstrated in The Master and his Emissary. It is not the what, but the how, that matters – as it does with the living person.

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We are constantly balancing, synthesising and alternating two entirely self-coherent, but on the face of it mutually incompatible, views of the world: we do it entirely unconsciously, and at the individual level most of us do it skilfully, in such a way as to respond appropriately to the richly diverse nature of experience. Understanding the hemispheric structure of the brain helps us see that, when the world view of a culture changes, it does not do so as an assortment of bafflingly independent changing elements, but according to a broadly coherent pattern that makes sense overall.

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Not ignorance, but ignorance of ignorance, is the death of knowledge. The right hemisphere, however, knows what it is that it must not get involved with.

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The phenomenological world of the left hemisphere is more self-directed, enclosed, self-validating – in thrall to its theory; the world of the right hemisphere more open to new information, the bigger picture, and what is actually the case, regardless of what the theory might suggest.

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many things are better understood by observing, not the ‘parts’, but the whole of which they are themselves part: seeing the Gestalt. I see metaphor as one aspect of the way we understand the world, something at which the right hemisphere is comprehensively better than the left,

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The truly foundational aspects of our experience – starting with consciousness itself – belong to the realm of things which permit no ultimate resolution. Indeed, some measure of our understanding of what we are dealing with is, precisely, the awareness that it forever exceeds our grasp. Not to be aware of the limits of our grasp of reality is folly, pure and simple. But, I emphasise, that does not mean that reality is whatever we happen to think it is, or would like it to be.

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In an age that prides itself on being capable of resolving and clarifying every aspect of experience in such a way as to explain it, in the hope of controlling it, we are too apt, when faced with a question that cannot be answered, either to deny that the question has meaning, or to deny that the problematic entity exists, or both. It is not just Zen wisdom that is founded on pondering irresoluble questions, and paradoxical injunctions: you scarcely need to be a Zen master to see the deficiencies in the all too common modern Western strategy of ruling questions impermissible, or denying the existence of what we can’t comprehend. The most important questions are, precisely, the unanswerable ones.

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Our dominant value – sometimes I fear our only value – has, very clearly, become that of power. This aligns us with a brain system, that of the left hemisphere, the raison d’être of which is to control and manipulate the world. The whole point of its knowing is, simply, the better to control: as Francis Bacon famously put it, knowledge is power. But for us to be satisfied that we have power, the world must be seen to respond to our will, and work according to rules that we can know and operate, whenever and to whatever extent our will dictates. Any suggestion that some things are not amenable to this approach is taken as an affront to the onward march of the human intellect. Odd as it may seem, it is the very same people that view humans as little more than conniving apes that are most vain about their human capacity to know and do. The position seems to be that, however limited, commonplace and unamazing we are, the world is even more limited, commonplace and unamazing, since with all our shortcomings we can understand it. This could be contrasted with the view that marvellous as we are, the world is more marvellous than we have the capacity to understand. As the British nuclear physicist Emerson Pugh put it: ‘If the human brain were so simple that we could understand it, we would be so simple that we couldn’t’.

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To quote Roger Sperry: Questions such as those concerning scientific truth, the nature of reality, and the place of man in the cosmos require for their study some knowledge of the constitution, quality, capacities and limitations of the human mind through which medium all such problems must be handled.2

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As the great English neurophysiologist Sir Charles Sherrington established more than a century ago, the proper working of the body depends on what he called ‘opponent processors’, systems that complement and counterbalance one another by their essentially opposing actions.

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The autonomic nervous system, which regulates body functions below the level of consciousness, is divided into the sympathetic system, which prepares the body for action (‘fight or flight’), and the parasympathetic system, which prepares it for repose: again, the finely modulated states that we require moment to moment are better carried out by opponent processors than one unidirectional system.

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the callosum remains by far the most immediate way in which the two hemispheres of the human brain interact, and it is the mediator not merely of connexion, but of hemisphere specialisation,

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More specialised units need to communicate less; with less communication, they become more specialised.

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And this, in turn, has an effect on the corpus callosum. It does not simply require a decrease, or an increase, in its importance. It requires a change of function. What it means is that the callosum, too, needs to differentiate and specialise. Conduction times short enough to support common activity in both hemispheres are possible in only a few instances, not in all, and must therefore be sparingly allocated.

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given specialisation and differentiation, one thing callosal tracts need to ensure is a sufficient degree of non-interference.30 And that means there must be mutual inhibition. A balance between imparting information, at the sensorimotor level, and inhibiting premature involvement, especially at the cognitive level, needs to be struck.

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The evolution of larger brain size in primates, not just in humans, results in increasingly independent hemispheres.36 This arrangement is more efficient for two main reasons: first, the avoidance of interference, especially in complex cognition; and second, a potential doubling of working capacity.

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Greater hemisphere independence is associated with superior performance: ‘importantly’, as one research group report, ‘the magnitude of lateralisation [of functions] … predicted the level of cognitive ability’.

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what is the frontal cortex for? Largely for stopping things happening.

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To members of a society in which doing things is much more highly valued than desisting from doing things this might seem paradoxical. Yet this negation is highly creative. It puts ‘necessary distance’ – in time and space – between ourselves and the immediacy of experience, so as to take the broader view, contextualise, test hypotheses and plan for the future. This enables us to stand back and see things in perspective, to choose whether or not to give in to the immediate reaction, or to see things in a different way, and respond more effectively; it enables us to delay gratification in order to achieve a greater end. It enables us to see our fellow creatures not merely as enemies but as potential allies; to see others, for the first time, as beings like ourselves, with needs and wants similar to our own. There is a proper distance between ourselves and the world which enables us to see what is going on: it

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apparent opposites (according to the linear, left hemisphere way of thinking), elements in sequence, often turn out to collaborate in fulfilling their ends. Things and their opposites depend on one another for their being, bring about something new through their opposition, and ultimately co-create the world. They are dipoles.

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With the advent of language, especially, being able to go offline creates a relatively abstracted and virtual world alongside the immediate world of perceptual experience.

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The most ancient difference between the two hemispheres lies in two kinds of attention. And so does the most highly evolved difference, enhanced by the frontal lobes. The more focal attention is narrowed, the more it takes its object out of the realm of time, space, the body and emotion. A virtual world is merely the conclusion of that process – and that is where the evolution of the left hemisphere leads: to a virtual world.

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Our world view is not simply the way we look at the world … world views create worlds. —Richard Tarnas1 The question is not what you look at, but what you see. —Henry Thoreau 2

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Attention is responsible not just for the how (though it is), not just for the what (though it is), but, as far as the left hemisphere is concerned, for the whether-at-all of existence. What it doesn't attend to, as far as the left hemisphere is concerned, doesn't exist.

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Left hemisphere attention is sharply restricted in space and time, which is, after all, the four -dimensional world in which we live. It tends towards precision, but at the expense of depth. It is no longer true to the expansive, always moving, always changing, endlessly interconnected nature of reality. One way of putting it is that the left hemisphere can provide some sorts of knowledge about the world, as it would be scrutinised from a certain theoretical point of view effectively outside the realms of space and time (as on a map); whereas the right hemisphere provides us with the knowledge of the world in space and time (as experienced). Constraining attention, left hemisphere fashion, makes coherent wholes dissolve, and brings things to a standstill: reality leaches out.

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The left hemisphere's limited appreciation of depth in space and time is in keeping with its tendency towards stasis. It seems to lack appreciation, not just of motion, but of emotion; it relatively lacks appropriate emotional depth, or concern, tending to be irritable or facetious, especially when challenged. It tends to disown problems, and pass the responsibility to others; is overconfident about what it cannot in the nature of things know much about; fabricates (often improbable) stories to cover its ignorance; sees parts at the expense of wholes; tends to see 'from the outside', rather than experience 'from the inside'; and has an affinity for the inanimate, and for tools and machines in particular. It is also quite confident it is right.

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Thinking is more interesting than knowing, but not than looking. —Johann Wolfgang von Goethe1 To repeat: don't think, but look! —Ludwig Wittgenstein2

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Perception is the act whereby we reach out from our cage of mental constructs to taste, smell, touch, hear and see the living world.

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In every sensory modality the right hemisphere appears to have an advantage over the left when it comes to primary perception. Perceptual distortions and hallucinations in each sensory modality are strongly associated with right hemisphere dysfunction. Thus, as far as both attention and perception go, the right hemisphere is a more important guide and a more reliable one to the nature of reality.

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Reasoning is classically associated with the left hemisphere, but in reality most studies show that both hemispheres contribute to reasoning; and the part played by the right hemisphere is significant, as we shall see.

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As I have repeatedly emphasised, the old dichotomy – left hemisphere rational, right hemisphere emotional – is profoundly mistaken, on both counts; not to mention the fact that reason and emotion are never entirely separable. Knowing the limits to reason is essential to understanding. If not coupled with contextual, implicit and intuitive understanding (in none of which the left hemisphere excels), it can magnify error.

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One ancient distinction in reasoning processes is that between induction and deduction. Induction is, in a more immediate sense than deduction, about predicting the future, and is based on observing the 'form' so far: the sun rose every morning to date, so it will in all likelihood do so tomorrow. Induction relies on the familiar remaining unchanged. This kind of reasoning is particularly associated with the left hemisphere, which makes sense if we realise that the left hemisphere needs to be able to predict, since its main concern is a plan of action. It therefore, unsurprisingly, likes predictability, familiarity and certainty. With that in mind, one has to be aware that this kind of reasoning is nonetheless intrinsically uncertain; the result is that the tendency of the left hemisphere is to treat things as more certain than they are . This can give rise to a problem. It could lead you to believe, for example, that all swans are white, simply because you have never seen a black one. Nassim Nicholas Taleb famously wrote a book, entitled The Black Swan, about problems of this kind: in his view they led to the Western economic collapse of 2008.

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Deduction, by contrast, is seeing that something is implied by what one knows, and is latent or implicit in it – though not yet explicitly stated. To take another example from the fauna of Australia: if mammals are those that suckle their young, and the platypus suckles her young, the implication is that the platypus is a mammal, even if she has a beak and lays eggs. Deduction is about uncovering and integrating latent meaning so long as it makes sense, and rejecting it if it is discrepant – but not just because it is unfamiliar, or necessitates a recasting of one’s thinking. If such a recasting seems reasonable, an advance in understanding has been achieved.

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Deduction is more persuasive than induction when deployed as an argument. For example, explaining that the mode of transmission of HIV makes it impossible to contract AIDS by casual contact (deduction) is more convincing than stating that no-one has yet caught AIDS from casual contact (induction).

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To put it crudely, the right hemisphere is our bullshit detector. It is better at avoiding nonsense when asked to believe it, but it is also better at avoiding falling prey to local prejudice and just dismissing rational argument because the argument does not happen to agree with that prejudice.265 And the left hemisphere? ‘The left hemisphere’, according to Ramachandran, ‘ is a conformist, largely indifferent to discrepancies, whereas the right hemisphere is the opposite: highly sensitive to perturbation.’266 Elsewhere he says: ‘the left hemisphere’s job is to create a model and maintain it at all costs’.267 Older subjects, as it turns out – sad news for those of us who have opted to grow older – exhibit a greater belief bias than younger subjects, and rely less on the right lateral prefrontal cortex.

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According to Michael Gazzaniga, the left hemisphere ‘is constantly looking for order and reason, even when there is none – which leads it continually to make mistakes. It tends to over-generalize, frequently constructing a potential past as opposed to a true one.’316 My model, says the left hemisphere, is better than your reality: in the canonical, theoretical world of general abstractions, things should work out according to my plan. But then reality, with all its messy complexities and differences, gets in the way and spoils the story.

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So-called ‘mirror neurones’ have become so topical that they have passed into popular culture , where there is a probably exaggerated idea of their significance. These are neurones that are active both in carrying out an action, and in watching another carry out an action. Clearly this is fascinating, because it suggests that we do literally experience some of what another is experiencing when we watch and empathise. Mirror neurones may lie behind the important capacity for imitation, which is an essential element in human evolution.68 But it is likely that such mirroring qualities are a feature of many much broader networks in the brain, and may simply describe an aspect of the functioning of the nervous system in general, whereby it attunes to other nervous systems, rather than of a special region within it.

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Social and emotional understanding are central to understanding all human situations. The evidence is that the right hemisphere is of critical importance for this, including the sense of reality itself, the ability to understand what another person knows, how that differs from what you know, what they mean and what their unspoken intentions might be. The right hemisphere is superior at emotional expression and receptivity. It is crucial for empathy and for a sense of agency. It is important for understanding implicit meaning, in all its forms, including metaphor, and for reading faces and body language. It understands how context changes meaning. In all these respects, the evidence is that it is superior to the left hemisphere .

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human intelligence is not like machine intelligence – modelled, as that is, on the serial procedures so typical of the left hemisphere. Comparing artificial intelligence with human intelligence, and indeed that of organisms more generally, the microbiologist Brian Ford writes that ‘to equate such data-rich digital operations with the infinite subtlety of life is absurd’, since intelligence in life operates on informational input that is essentially Gestalt and not digital. [Living systems] can construct conceptual structures out of non-digital interactions rather than the obligatory digitized processes to which binary information computing is confined.27

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Evidence from a number of sources suggests that the right hemisphere contributes the majority, not just of emotional and social intelligence, but also of what is ordinarily meant by intelligence (IQ) – cognitive power, or g. This appears to be particularly true among children and adults of the highest intelligence.

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This shape – to put it in banal terms, trying, failing, relaxing and then succeeding – is a fairly good way of understanding the creative process. But the fact that it depends on unconscious processes, flexibility, openness to the unknown and undetermined, relinquishing control, making imaginative leaps, seeing analogies, metaphors and images – all elements that are more characteristic of the right hemisphere than the left – does not endear it to a certain type of science practitioner who likes his or her world left hemisphere fashion: neat, predictable, and determined by explicit procedures that are known and can be followed whenever one wishes. Mechanical and under control, in other words; rather than messily alive and unpredictable.

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The key element in generation seems to be the ability to think of many diverse ideas quickly, demanding breadth, flexibility and analogical thinking – seeing likeness within apparent dissimilarity. This could be, and often is, summed up as ‘divergent thinking’, a term that is deliberately (and appropriately enough) loose, but suggests the bringing together of non-adjacent ideas, adaptability of thinking and originality of approach. According to a recent meta-analysis, the largest to date, of the correlates of creativity in the brain, ‘the hallmark of creativity is divergent thinking’.

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What divergent thinking covers is not just being able to make up new ideas at random – most of which would be worthless – but perceiving connexions and shapes or forms that guide thinking by analogy: to broaden a field that has become too narrow, or to find alternative ways of visualising something that has become too familiar. All the literature on creativity in whatever field makes the same point: that it is about seeing hitherto unperceived parallels, seeing shapes or Gestalten that others have failed to see, standing back and taking the broad view, not squinting at the same microscopic field and looking up the rule book. Talent hits a target no-one else can hit, wrote Schopenhauer; genius hits a target no-one else can see.

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Creativity involves the making of hitherto concealed connexions. As Henri Poincaré noted, creative ideas are those which reveal to us unsuspected kinship between other facts, long known, but wrongly believed to be strangers to one another. Among chosen combinations the most fertile will often be those formed of elements drawn from domains which are far apart.

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The broader one’s understanding of the human experience, the better design we will have.

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Linear approaches and analytic thinking, characteristic of the left hemisphere, are fine in the right context, and may at a subsequent phase take part in creativity by narrowing things down and eliminating some of them, but on their own will not achieve creativity.

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Interestingly, while the importance of a broad view is obvious when conceived in spatial terms, we forget that it also involves a temporal element: integration of ideas from different periods of one’s life, or even from different eras of history.

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For creativity to succeed, then, there needs to be breadth of vision; the capacity to forge distant links; flexibility rather than rigidity; a willingness to respond to a changed or changing context; as well as tolerance of ambiguity and of knowledge that is, at least at the outset, inherently imprecise.

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creativity escalates with still higher intelligence: creative achievement benefits from higher intelligence even at fairly high levels of intellectual ability … This is in line with previous studies reporting that IQ is predictive of creative achievement even within high ability groups.

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the more you understand creativity, the less you will be inclined to imagine there are techniques for achieving it, even for those with any degree of aptitude. We can't make creativity happen, but we can certainly do our best to stand in its way. Or therefore not. Furthering creativity is mainly about not doing, rather than doing. And in particular not doing any of the things that look obvious: trying hard, trying hard not to try, trying a systematic approach, trying a random approach, and so on.

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Preconceptions hamper the generation of solutions since they easily turn into constraints. 27 Consequently, one of the ways to encourage flexibility is the relaxation of self-imposed constraints.28 And, most importantly, the adoption of the stance of someone who does not know, and is prepared to listen.

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Thus in creativity there are bound to be at least, and at the very simplest, two phases, that may often nonetheless overlap: a phase of exploration guided by the imagination; followed, at a distance of anything from seconds to months, but in any case followed, by a critical sifting and evaluation of the results. Analytic, critical, convergent thinking must have something to work on: you can't think critically without something to criticise, or narrow down the field without having a field to narrow.

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It is by logic that we prove, but by intuition that we discover. To know how to criticize is good, but to know how to create is better … Logic teaches us that on such and such a road we are sure of not meeting an obstacle; it does not tell us which is the road that leads to the desired end. For this it is necessary to see the end from afar, and the faculty which teaches us to see is intuition.

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Gestalt perception, on the other hand, when based on a sufficient wealth of unbiased observation, has a way of being right, and if one is familiar with its occasional trick of being altogether wrong and knows when to discount its assertions, it is an invaluable and quite indispensable guide.

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Indeed creativity, sadness, non-verbal thinking, empathy, intelligence and a preponderance of right hemisphere activity are all multiply interconnected.

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Since we have noted that thinking in images, models and analogies is a prerequisite of creativity, while thinking in words is not (and may actually inhibit creativity at the truly generative stage, rather than at the later more consciously critical stage), it might be expected that mood changes that promote the right frontal cortex will be associated with creativity in a way that those engaging primarily the left frontal cortex will not.

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The economist and statistician Karol Jan Borowiecki used linguistic analysis software to scan the text of 1,400 letters written by Mozart, Beethoven and Liszt to their friends, colleagues and loved ones, and analysed the results for the expression of positive and negative emotions, such as joy, love, grief and heartache.288 He then plotted the course of mood against creative production. The resulting curves are striking: he found a consistent link between the intensity of negative emotions, especially sadness, and subsequent periods of artistic brilliance. He concluded that such sadness was not just correlated with creativity but that it actually had a reliable causal effect, typically delayed by some months.

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study showed that 'normal scientists', those that are concerned with preserving the paradigm, and 'revolutionary scientists' whose work transcends the paradigm, have quite different profiles. While the paradigm-preserving, 'normal', scientists had lower than average levels of mental illness, the paradigm-rejecting, 'revolutionary' scientists had higher than average levels – in this respect resembling poets and artists.312

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Creativity involves a number of elements in which the right hemisphere is superior to the left: breadth of vision, the capacity to forge distant links, flexibility rather than rigidity, a willingness to respond to a changed, or changing, context, a tolerance of ambiguity, and an ability to work with knowledge that is, for the most part, inherently both imprecise and implicit. It involves analogical thinking and visualisation; it is often expressed in music, metaphorical and original language, and visuo-spatial imagery, all of which are also better served by the right hemisphere than the left. It can easily be inhibited by too close following of rules, too familiar patterns of thought, a requirement for certainty and clarity, and a concern for detail too early in the process, all of which tend to be favoured by the left hemisphere. Thus, though the left hemisphere has a role to play during the later implementation of a creative insight – what I call the translational phase – it tends to impede during the generational and permissive phases, which correspond to what most people mean by creativity. There seems to be a marked increase in reliance on the right hemisphere among the highly creative, compared with the averagely creative, and not observing this distinction in experimental design has sometimes muddied the waters of research into hemispheric differences. Nonetheless, there is a very large body of evidence suggesting that the right hemisphere plays the key role in creativity.

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'Illness is philosophically revealing and can be used to explore human experience', writes the philosopher Havi Carel: I suggest that illness is a limit case of embodied experience. By pushing embodied experience to its limit, illness sheds light on normal experience, revealing its ordinary and therefore overlooked structure. Illness produces a distancing effect, which allows us to observe normal human behaviour and cognition via their pathological counterpart . I suggest that these characteristics warrant illness a philosophical role that has not been articulated.

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Is it conceivable that elements of the way of being in the world embedded in a culture might start to resemble one hitherto largely confined to the mentally ill?

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Might it be, then, that as a culture we were exemplifying not, of course, a sudden epidemic of schizophrenia, but too heavy a reliance on the world as delivered to us by the left hemisphere, meanwhile dismissing what it is that the right hemisphere knows and could help us understand?

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Western modernity has many overlapping features with the phenomenology of schizophrenia, as Louis Sass has convincingly demonstrated in Madness and Modernism; and I submit that this is because modernity simulates not a disease state, but a hemispheric imbalance, as I suggested in The Master and his Emissary. The testimony of people with schizophrenia and autism provides us with articulate accounts of what living in the left hemisphere's world feels like, when most fully expressed. Such accounts help to make sense of apparently disparate phenomena around us, and give us a perspective on our unexamined assumptions about the fabric of reality

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the left hemisphere is, compared with the right hemisphere, unreliable in just about every way that matters. In terms of attention to the world, and its role in thereby constructing, and understanding, experience; in its inability to comprehend time, space and motion; in its lack of skill in conveying and interpreting emotion; in its (lack of a) sense of the body as a living inseparable part of the self; in the comparative weakness of its faculties for direct perception, for the evaluation of beliefs and for making judgments; and indeed in terms of its lesser intelligence (which means understanding): in all of these it is more vulnerable to falsehood, more likely to deceive us, than the right.

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The right hemisphere-damaged subject, overly committed to the left hemisphere mode of simply putting together the pieces, joining the dots, and following procedures, is unable to appreciate that the picture he arrives at is bizarre and incoherent. He placidly accepts the preposterous nature of his conclusions, and, if things seem improbable or awry, he no longer feels the strange or surprising nature of them. They fail comprehensively to alert him, in the way that they normally should, that there is an aberration. In this he is like schizophrenic subjects, who adopt what Sass & Byrom call an 'anything goes' orientation, in which they ' quickly identify, accept and take in stride phenomena that most people would find anomalous, strange to the point of being difficult to recognize'.

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The left hemisphere's world is now an increasingly virtual world. It no longer even pretends to yield a faithful portrayal of reality. For that it depends on the right hemisphere. It has, it thinks, more important – certainly other – things to do. It is there to unpack the implicit in what it is given about the outside world, make it explicit and deal with it according to the rules. It is there to aid strategy. Unfortunately, by being purely strategic in intent, the left hemisphere makes strategic mistakes, since it remains largely ignorant of the reality on which it relies. As a sophisticated computer would. And very soon, no doubt, will.

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When you are out of touch with reality you will easily embrace a delusion, and equally put in doubt the most basic elements of existence. If this reminds you of the mindset of the present day materialist science and philosophy establishments, as well as of the loudest voices in the socio-political debate, we should not be particularly surprised, since they show all the signs of attending with the left hemisphere alone. I live in the hope that that may soon change: for without a change we are lost.

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the aspect of reality revealed by the left hemisphere must be contextualised by being taken up into the broader, and deeper, overarching vision available to the right hemisphere; alas, when the left hemisphere takes the lead, it leads us astray. As always, it is a good servant but a poor master.

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The single most profound difference between the hemispheres, which I will have cause to return to repeatedly, is the distinction between the experience of something as it 'presences' to us in the right hemisphere, and as it is 're-presented' to us in the left.

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The whole of Merleau-Ponty's philosophy of embodied understanding depends, in one way or another, on a fundamental distinction: that between two modes of encountering the world, which he calls 'je pense' and 'je peux'.2 The first deals with a mental representation in which we are quite separate from the world, and the question is, then, how our thoughts can be said to relate to that world; the second is an intuitive embodied awareness of the world in which we are already fully situated, a field of potential for interaction, something that comes prior to any conscious re-presentation.

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We now tend to think of truth as a matter of propositions. The word 'truth' in its origin indicates not a proposition, but a disposition. 'True' (cf German treu, faithful) is related to ' trust,' and is fundamentally a matter of what one believes to be the case. Truth and trust (belief) go together. One cannot have trust in a society where there is no truth; and one cannot be true to a society in which there is no trust. That is of fundamental importance, since, as Confucius told his disciple Tzu-kung, for a stable society a ruler needs three things: weapons, food and trust. If he cannot hold all three, he should forgo weapons first, and food next; for ' without trust we cannot stand'.

9,528

One of the most ignorant remarks ever made, representing the machine world view for which we are now paying, and will continue to pay, a very high price, is Henry Ford's 'history is bunk'.

9,548

The idea of truth always implies some sort of correspondence. One major strand in thinking over the centuries is, indeed, known as 'correspondence theory': that truth lies in a correspondence between what goes on in our minds and a mind-independent reality. The closer the two come, the more we can speak of truth.

9,555

The classical counterpoint to correspondence theory is 'coherence theory': that truth lies in the cohesion of what it is that goes on in our minds. If something coheres with – and, moreover, mutually sustains – other things that we hold to be true, it may be regarded as itself true.

9,601

Truth is uncertain not because it is empty, but because it is full – rich, complex, manifold. (This is related to what I believe to be the mistaken assumption that because we cannot pin down the 'meaning' of the world in which we live, it has no meaning:

9,613

If we place the emphasis on truth as a relationship, which is where we started, rather than as a 'thing', certain aspects become apparent. In a relationship, both parties count, and the relationship isn't a separate 'thing', but contains both parties that constitute it. It doesn't lie in one or the other alone. This means that it is necessarily lived; changing and contextual.

9,652

Note that it is the precision of Newtonian mechanics that makes it a less good fit, while the intrinsically less precise quantum mechanics is more accurate. I am reminded of the way in which template-driven 'packages of care', now enjoined as good practice by managers of our healthcare system, are always, despite their crispness, a poor fit to the individual case because general, and precise – whereas the fuzzier, personally tailored response is a much better fit, because individual, and accurate.

9,690

The most immediately concrete way of understanding this non-dichotomy is in human relationships. A good relationship is one in which each party is maximally fulfilled as a differentiated individual, without this in any way detracting from the relationship, but rather being a condition of the possibility of a good relationship at all – and indeed also a consequence of such a relationship. Equally, we are not atomistic, and cannot be 'ourselves', whatever that would mean, in a cultural or societal vacuum. Thus a good society is not one in which individuality is lost, but one in which it is fulfilled; yet, at the same time, that individuality must not be a threat to the cohesion of the society. There is such a thing as tyranny of individuals over society, as well as society over individuals.

9,727

For, as human beings, we must inevitably see the universe from a centre lying within ourselves and speak about it in terms of a human language shaped by the exigencies of human intercourse. Any attempt rigorously to eliminate our human perspective from our picture of the world must lead to absurdity.

9,730

All the same, we could make an imaginative attempt to stand aside from our usual viewpoint. Indeed, one of the crucial functions of the imagination is to enable us to take various perspectives on the world: it enables us to 'see how it looks from over there'. This encourages a certain humility about how it looks from one's own perspective, and yet at the same time provides some reason for confidence in what one takes to be the case. Without imagination there can be no objectivity.

9,735

Returning to the problem of the misconception of the relationship of the hemispheres as oppositional, this is as simple as whether one takes a left hemisphere, or right hemisphere, view of opposition: one point of view, that of the primarily analytic left hemisphere, is exclusive ('either/or'), while the other, that of the primarily synthetic right hemisphere, is inclusive ('both/and'). Since this is their disposition in general, it also applies to their relationship in particular. To recall the primary myth of the Master (RH) and his emissary (LH), the Master knows he needs the emissary, while the emissary does not know that he needs the Master. The tendency is for the one that sees less (the emissary) to believe that he sees all, while the one that sees more (the Master) sees there are things he doesn't know; just as those who think they know it all know less than those who know they do not.22 To the extent that there is a 'war' here, it is of the left hemisphere's making, and stems from its unwarranted self-belief.

9,767

We have become accustomed, in our Western way, to see all exchange of ideas as a kind of battleground in which there will be conquest and defeat. In the East, ancient truths have not commonly been thought such that they can convince (the word comes from Latin vincere, to conquer) through argument, but rather that they become appreciated (the word comes from Latin pretium, value) through a patient, disciplined opening up of the self to experience. Jan Zwicky puts it like this: Truth is the asymptotic limit of sensitive attempts to be responsible to our actual experience of the world … 'sensitive attempts to be responsible' means truth is the result of attention. (As opposed to inspection.) Of looking informed by love. Of really looking .

9,778

Though truth is always my personal judgment, it is not just possible, but necessary, that my judgment should take into account yours and many others. It is far from random, but is, rather , informed by experiment, perception, reason, intuition and imagination. That doesn't make it less reliable than being informed by a single source, such as reason, might have done, but more reliable. Acquiring a degree of judgment that can make these elements intelligently cohere is – or used to be – the whole purpose of education. It is why we study the humanities. What history and classics and literature tell us is not to be found in the sciences anywhere.

9,783

Judgment used to be the foundation of the idea of reasonableness – a concept you may remember, but which we are in danger of losing, if we have not already done so, in a mechanised, bureaucratic society. The popular reaction to this has been only to intensify the mechanistic vision: no longer seeing complex, unique individuals but only representatives of groups, no longer open to appropriately nuanced, but simple 'I'm right, you're wrong' positions, and shouting more and more loudly. Reasonableness is as far from unbridled emotion as it is from rote rationality, on the worst excesses of which it acts as a much-needed brake.

9,878

both the views currently on offer in so much public debate today – naïve positivism and naïve deconstructionism – are typical left hemisphere fictions.

9,882

That there is ultimately no certainty (except in mathematics) is emphatically not an invitation to promulgate your latest theory based on some political ideology: the dangers in this are every bit as great as in adhering too closely to the reductionist materialist picture – possibly greater, since they derive from a position for which every point of view, every statement, is as valid as any other. That brings an end to truth, and hence to civilised discourse: indeed, an end to civilisation. Unfortunately, naïve positivism may ultimately lead there, too.

9,909

The antagonism between science and metaphysics has, like all family quarrels, been disastrous. —Alfred North Whitehead

9,932

science cannot possibly fulfil the burdensome role of sole purveyor of truth. This is not a failing in science. Good science is aware of its limitations. Scientism, the belief that science will one day answer all our questions, is not.

9,966

Explanation, science's forte, is a subset – an explicit, rigorous, disciplined subset, but still a subset – of understanding. All understanding depends on metaphor. What we mean when we say we understand something is that we see it is like something else of which we are already prepared to say 'I understand that'. That, in turn, we will have understood because we have likened it to something else we had previously understood, and so on. It's metaphors all the way down.

10,036

we cannot discover whatever aspect of truth is revealed by a certain approach unless we commit ourselves to it sufficiently to find out; but by the very act of doing so, other aspects of truth become concealed, aspects which, when pointed out by others, will tend to be unrecognised, or ultimately dismissed.

10,122

[Michael Polányi] notes that all methods, even the most formalised, rely upon our judgements . It is not possible to absent ourselves, and nor should we try. If all knowing involves the participation of a knower, then it follows that all knowing is a form of valuation. We make judgements about what we ought to believe. Our beliefs may be mistaken, but they are not arbitrary. They are guided by our tacit awareness. In his analysis of the structure of tacit knowing – his key contribution to philosophy – Polányi seeks to reconcile fact and value (which in the modern period have been broken apart).18 Just because we can't measure values, they are neither less important, nor less rightly operative, in science: the confusion comes from science's reluctance to accept as real what cannot be pinned down and measured. It therefore unwittingly drives itself into a cul-de-sac: the false divorce of fact from value.

10,152

if anything goes, nothing goes: we no longer have any purchase on reality. My position is that almost nothing goes: the whole purpose of education, self-discipline and scholarship being the honest, open and humble attempt to find whatever lies in that space closest to the truth, in the knowledge that success is always a matter of degree. But that degree is of the utmost significance.

10,167

Whitehead: 'There are no whole truths; all truths are half-truths. It is trying to treat them as whole truths that plays the devil.'22 And, with particular relevance to science: In scientific investigations the question, True or False?, is usually irrelevant. The important question is, In what circumstances is this formula true, and in what circumstances is it false?

10,180

There is among many of the public, as Nassim Nicholas Taleb points out, 'a religious belief in the unconditional power of organised science, one that has replaced unconditional religious belief in organised religion …We have managed to transfer religious belief into gullibility for whatever can masquerade as science.'26 No good is done to science by making the inflated claims that, unfortunately, some naïve scientists do make on her behalf – thinking that they thereby do her a favour. Obvious discrepancies between the excessive claims of any culture and human experience lead eventually to a countermovement that ends in disaster: the baby gets thrown out with the bathwater.

10,253

John von Neumann put it: The sciences do not try to explain, they hardly even try to interpret, they mainly make models … The justification of such a mathematical construct is solely and precisely that it is expected to work.38

10,266

According to the myth, scientific knowledge arises from the rigorous, logically sequential unfolding of whatever follows securely from predictable procedures by calculation according to fixed laws. But this myth utterly fails to do justice to what the work of science actually involves. The sheer, staggering work of imagination and inspiration that is science, its discoveries unpredictable and recalcitrant to compulsion by laws or procedures – real, living science – is swept away in this myth, this representation of science as something pedestrian, uninspired, essentially bureaucratic – and dead.

10,283

you can't even start on the scientific method without making assumptions, using judgment and having insights, insights that are not part of the method, and do not follow any kind of formal principles – and the method is in any case unlikely to produce 'important' results.

10,443

Science understandably does not find a purpose, because purpose is not something its way of thinking recognises. That is not a criticism of science per se. At most it is a criticism of a certain 'philosophy' of science, rarely critically examined, or even challenged; more generally it is a criticism of those who want to take the methodology of science into realms where it has no purchase on reality. When it comes to the really big questions – such as the nature, meaning or purpose of life, consciousness, time, love, energy, matter, beauty, truth or goodness – science encounters the limitations inherent in its dependence on models as the necessary means of proceeding, for in these cases there not only isn't an appropriate model, there can't be one, because each of them is completely sui generis.

10,662

As David Bohm commented in the 1960s, it is an odd fact that, just when physics was moving away from mechanism, biology and psychology were moving closer to it. 'If the trend continues', he wrote, 'scientists will be regarding living and intelligent beings as mechanical, while they suppose that inanimate matter is too complex and subtle to fit into the limited categories of mechanism.'9 He was not mistaken.

10,988

change in a living organism takes place on an unimaginable scale. Let me give some idea of that scale. There are an estimated 37.2 trillion cells in the human body.64 Each one of these cells performs many millions of complex reactions every second.65 In doing so the cell does not act atomistically but within complex feedback systems with other cells. Biological enzymes promote extraordinary rates of change. In the absence of catalytic enzymes, the decarboxylation of amino acids would proceed with a typical half-life of about a billion years: in the presence of enzymes these half-lives are reduced to less than a thousandth of a second.66 And a single molecule such as carbonic anhydrase – of which the body contains unimaginable numbers – can catalyse upwards of a million reactions per second.67 Each protein may combine with 'several hundred different modifier molecules, leading to practically infinite combinatorial possibilities' and each 'protein itself is an infinitesimal point in the vast heaving and churning molecular sea of continual exchange that is the cell'.

11,058

in biological systems, causation tends to follow not straight lines, but spirals, involving recursive loops, and multiple causes leading to multiple effects across a network, with sometimes competing factors cross-regulating one another, reciprocally interacting, and in ways we do not understand taking information from the whole. All this makes the classic idea of a two-element, cause and effect sequence unhelpful.

11,065

'the heart of the problem lies in the fact that we are dealing not with a chain of causation but with a network', something like a spider's web, in which a perturbation at any point of the web changes the tension of every fibre in it.81 Context is everything. Which means we cannot simply account for organisms from the bottom up, but must do so at least as much from the top down. Not to mention from the sides, as in a web. We might be accustomed to thinking of biological processes in the abstract, isolated artificially by our mode of attention. But each element in each process is likely to be involved in several other 'causative chains', more like an unimaginably complex piece of choreography, where members of one group pass in and through another, for a while belonging to both or neither – each process having its own apparent end.

11,118

The entire universe must, on a very accurate level, be regarded as a single indivisible unit in which separate parts appear as idealisations permissible only on a classical level of accuracy of description. This means that the view of the world being analogous to a huge machine, the predominant view from the sixteenth to nineteenth centuries, is now shown to be only approximately correct. The underlying structure of matter, however, is not mechanical … This means that the term 'quantum mechanics' is very much of a misnomer. It should, perhaps, be called 'quantum nonmechanics'.

11,707

A machine's functioning is linear, whereas the process of living things is of entities that mutually co-arise.

11,866

To move from a focus on things as the ultimately important constituents of reality, to one in which processes are the important entities, requires a shift in our perception of space. We relinquish the idea of causation as strictly 'bottom up', from parts to wholes: we come to accept that whole systems contribute to the properties of their parts.228 The first is the only one that makes sense to the left hemisphere; in its view the whole can be understood only if we understand the parts. It takes the right hemisphere to understand that the parts can be understood only in the light of the whole.

12,088

The words for Nature in Chinese, tzu-jan (ziran), and in Japanese, shizen, mean whatever is ' of itself', exists 'spontaneously', is 'just what it is'. They are, in origin, adverbs, not nouns – ways of being, not things.269 If there is anything in this ancient perception, and I believe there is great wisdom in it, a vision of the natural world as a thing, and a mechanical one at that, is bound to restrict our understanding of what we are dealing with to a certain rather alienating perspective. A machine implies existence of an external creating force with its own purpose: Nature delights in her own.

12,211

At a fundamental level is nature discrete or continuous? I see no evidence whatever for discreteness. All the discreteness we see in the world is something which emerges from an underlying continuum … Quanta are emergent … they are not built into the heart of Nature.

12,273

Our attempts to view biological systems as mechanical–materialistic machines have failed dismally.

12,360

Mechanism is a perfectly useful way of looking at tiny details in a complex picture so as to be in a position to manipulate them. There it has had spectacular successes. But we mustn't be blinded by them. The problem is thinking that the same thinking will help you understand the whole: it can't.

12,372

'An organism is less like a machine than it is like a language whose elements … take unique meaning from their context'. Analyses of individual words and their possibilities of meaning can be an essential first step: without a knowledge of the words, we cannot grasp the whole. But at the same time, it is only the meaning of the whole that gives the individual words their full and proper significance.

12,383

Most human institutions, by the purely technical and professional manner in which they come to be administered, end by becoming obstacles to the very purposes which their founders had in view. —William James 1

12,515

A book gives time and space for ideas to be properly developed. The scientific paper, in particular the conventional contemporary science paper, is best suited to the dissemination of data, rather less well to the exposition of seminal ideas, though it does happen. Nowadays career scientists don't write books. Books have no career value, because they have no 'impact factor' associated with them, and they take (or should take) a long time to write, time which could be more usefully applied to producing papers. The parts are more valuable than the whole:

13,134

Originality, branching away from the conventional path, challenging received wisdom, is the essence of good science. Without it, science fossilises: it loses intellectual power, and purchase on reality.

13,359

Richard Feynman said during his Nobel Prize acceptance speech in 1965: ‘A very great deal more truth can become known than can be proven.’ And, I might add, a very great deal more falsehood can become known than can be disproved.

13,379

Some people think that science must be a principally left hemisphere business. I’d caution that, like absolutely everything else, it has its left hemisphere and its right hemisphere aspects. And, like everything else, science profits from balance and harmony between them. Some kinds of scientific thought, especially the drive to see the overall shape, to draw together disparate findings, to make connexions, to achieve imaginative insight, to evolve, or to modify a hypothesis, to observe things as near as possible as they are (empiricism), and to see whole systems that can’t be reduced to linear connexions, are likely to be more right hemisphere-dependent; the drive to make fine distinctions, to investigate localised areas or functions, to preserve the theoretical paradigm at all costs, to achieve certainty, to see the world in terms of things that are cause and effect to other things, in predictable, linear fashion – all this is more likely to express the contribution of the left hemisphere. Both are essential: though here as always, it is the right hemisphere’s role that needs to predominate, taking under its aegis the contribution of the left hemisphere. There needs to be a balance between opening up the flow of thought to possibility and closing it down towards an always provisional certainty.

13,435

I believe some of the political difficulties we are now experiencing in the Western world come from a lack of balance, in which wilful blindness to the problems in one’s own position eventually causes an ignorant and extreme reaction, when those who are at some level all too aware of what is being ignored and denied can finally contain themselves no longer. Those liberal intellectuals who had their doubts about the Western liberal intellectual consensus of the last few decades were stigmatised for not having unlimited enthusiasm. Those who criticised the consensus liberal position for its excessive one-sidedness, before it was too late, were seen as enemies: they were in fact friends. ‘Plato is my friend, but a greater friend is Truth’, as Aristotle is said to have remarked; falsely, as it happens – but such is life.

13,456

It is equally excessive to shut reason out and to let nothing else in. —Blaise Pascal

13,490

English has, for better or worse, become the lingua franca of the modern world. The distinction I am making between two kinds of reason is disguised in English by its using the same word for two very different phenomena. In other languages, it is not necessarily so. In German there is a distinction between Verstand and Vernunft,

13,495

I suggest that these two versions reflect the typical mode of operation, and exemplify the characteristics of, the left and right hemispheres, respectively. The first is rigid, aims for certainty, tends to ‘either/or’ thinking, is abstract and generalised, ignores context and aims to free itself from all that is embodied, in order to gain what it conceives to be eternal truths. The second is deeper and richer, more flexible and tentative, more modest, aware of the impossibility of certainty, open to polyvalent meaning, respecting context and embodiment, and holding that while rational processing is important, it needs to be combined with other ways of intelligently understanding the world.

13,517

And it is one of the messages of this book that imagination is not an impediment, but, on the contrary, a necessity for true knowledge of the world, for true understanding, and for that neglected goal of human life, wisdom.

13,564

No ‘-ism’ that is already parti pris can offer a ‘rich and ordered landscape’, because the mind that gives rise to it is closed, disregards the whole tapestry of reality in favour of just one strand, and inevitably needs to disparage those points of view it doesn't share. The rich view, by contrast, will draw from a number of standpoints and achieve a balanced synthesis.

13,573

Reasoning in the sense of rationality is a consistency tool, and nothing more. Its job is to tell us that if we hold a and b, and if we also believe that logical contradiction necessarily implies falsehood, then we must hold c as a consequence. It is an intermediate processor, albeit a very valuable one. It cannot ground itself, at the 'bottom' end, or give meaning to its outcome at the 'top' end. At the bottom end it must rely on axioms assumed on the basis that they are intuitively true (the word 'axiom' comes from the Greek word axia, meaning 'value' – in other words, reason is founded on what are in essence value judgments). Reason, like both science and, in a different way, morality, can bring a verdict into doubt only if it is grounded on something which is exempted from the doubting process.

13,623

The linear path is seductive to the left hemisphere's mindset in its directness and simplicity. But what seems like a virtue – that it does not change direction – reveals itself to be a vice. It has deep rhetorical power over Western minds, making it easier to believe not only that the direct approach is always the most rational, which it is not, and that reciprocal, contextual and feedback effects can be ignored, which they cannot, but that if something is good, twice as much of it will be twice as good, which is rarely the case, and that apparent opposites cannot coincide, which they can.

13,691

much Western philosophy has suffered from a ratiocentric bias – the notion that calm and detached rational analysis provides the unique key to understanding ourselves and our activities. At its worst, ratiocentrism involves a fantasy of command and control, as if by sufficiently careful use of reason we could gain an exhaustive understanding of the human condition.

13,727

the combination of scientism and ratiocentrism is making our vision narrower than it should be. I commented that, inside the bubble (which I would identify with the left hemisphere's reality) we no longer see the limitations of our viewpoint – precisely because of its limitations . But nothing can prevent us from noticing that things are not working out the way we hoped.

13,850

the way each of us makes sense of who we are and our relation to the world is a fearsomely complex process of which our intellectualising is only the thinnest of surfaces. At the very least this suggests the need for a certain humility about the philosophical project of 'making sense of ourselves and our activities'. We need to recognise the limitations of intellectual analysis, and the way in which insight is achieved not just by the controlling intellect, fussily classifying and cataloguing the pieces of the jigsaw, but by a process of attunement, whereby we allow different levels of understanding and awareness to coalesce, until a picture of the whole begins to emerge.

13,960

The lack of a rational understanding of the limits to reason may prove fatal. The coherence of a civilisation depends on accepting the reality and value of principles which we do not, and perhaps never can, fully understand. In all societies hitherto this has been achieved by the influence of knowledge embodied in traditions, sometimes religious, that asked for acceptance by an appeal to imagination, not to reason alone. 'The most dangerous stage in the growth of civilization', wrote Friedrich Hayek, may well be that in which man has come to regard all these beliefs as superstitions and refuses to accept or to submit to anything which he does not rationally understand. The rationalist whose reason is not sufficient to teach him those limitations of the powers of conscious reason, and who despises all the institutions and customs which have not been consciously designed, would thus become the destroyer of the civilization built upon them.

13,972

What secular reason is missing is self-awareness. It is 'unenlightened about itself' in the sense that it has within itself no mechanism for questioning the products and conclusions of its formal, procedural entailments and experiments.

13,975

Pascal, one of the greatest philosopher-mathematicians that ever lived, nonetheless said that ' the ultimate achievement of reason is to recognize that there is an infinity of things which surpass it. It is indeed feeble if it can't get as far as understanding that.'72 Kant demonstrated the limitations of reason in what are called his four 'antinomies' (or paradoxes): in each case reasoning about a fundamental aspect of reality leads equally, he claimed, to a conclusion and to its opposite. In the last century Kurt Gödel demonstrated definitively that every formal system is incomplete, in that each leads to conclusions that can be shown to be theorems of the system, but cannot be proven within the system. Mathematician and philosopher Greg Chaitin claims that 'an infinite number of true mathematical theorems exist that cannot be proved from any finite system of axioms'.73 It is not rational to assume, without evidence, that rationality can disclose everything about the world, just because it can lead us to the disclosure of many important elements.

13,998

Of course one does not need to rely on abstruse mathematics to see the limitations of reason. Anyone who understands poetry, drama, ritual, narrative, music, painting, architecture, or the sheer beauty and majesty of the natural world – or for that matter has ever fallen in love – can see that ultimate meaning will always lie beyond what reason can conceive or everyday language express.

14,020

Ambiguity is of the essence of human existence, and everything we live or think has always several meanings. —Maurice Merleau-Ponty

14,038

Music is as near an example of pure form as one might find: it does not refer outside itself. And yet its meaning is profound. Meaning arises richly and inevitably out of any embodied form: but it must be embodied. Only in algebra, which is disembodied, can form and meaning be separated. Frege's 'concept notation' is a kind of algebra which respects logical structures, but is indifferent to meaning. This, then, is the ultimate triumph of decontextualisation and abstraction.

14,231

All abstractions are only our own simplifications of something that came to us originally in the flesh and blood as perceiving, sensate, intuiting beings, enmeshed in an unfathomable network of human society and its history. 'Abstraction', according to John Kay, 'is the process of turning complex problems we cannot completely describe into simpler ones that we think we can solve. But gauging which simplification is appropriate requires judgment and experience. Our simplifications are idiosyncratic and subjective.'36 And in testing certain features of human decision-making, the circumstances have to be altered and simplified to yield artificially clear objectives, divorced from any context. In doing so the model-makers ' are substituting a problem their methods can solve for the problem we actually face'.

14,240

'The belief that models are not just useful tools but are capable of yielding comprehensive and universal descriptions of the world blinded proponents to realities that had been staring them in the face', wrote Kay of leading Western economists prior to the crash of 2008: That blindness made a big contribution to our present crisis, and conditions our confused responses to it. Economists – in government agencies as well as universities – were obsessively playing Grand Theft Auto while the world around them was falling apart.'

14,315

There is a kind and a degree of reaching after precision, clarity and rigour which is misplaced, because its subject becomes more and more tenuous as this process continues. It ends in a kind of increasingly unrewarding pedantry, reminiscent of an anorexic's attempt to split a pea rather than swallow it. The attempt is to over-clarify an area that intrinsically does not permit it. 'It is the mark of an educated man', wrote Aristotle, 'to look for precision in each class of things just so far as the nature of the subject admits'.

14,320

Human affairs are a case in point. Knowledge of something that is by its nature not precise will itself have to be imprecise, if it is to be accurate.

14,365

The right hemisphere activation is at the same time more provisional, less prepotent, than that of the left hemisphere, and more diffuse, engaging a much broader semantic field, including elements only distantly related.54 The point to understand here is this: the less sharply each word's meaning is specified, the more likely it is to connect to other words and concepts. Needless to say, this makes the right hemisphere invaluable in enabling us to understand a range of implications,55 to see the overall 'point',56 to understand all that is non-literal in language,57 and to gain insight into an interrelated whole.

14,404

There is an important role for ambiguity (polysemy) in all understanding, something which metaphor, symbol, humour, tone of voice, and all the implicit modes of expression of the arts best understood by the right hemisphere, imply. This is not ambiguity in the sense of just not having taken proper trouble to pin down the meaning: but in the sense of having, indeed, taken proper trouble not to pin down the meaning too closely, so as to let it live – the proper rigour of ambiguity.62 Bringing an unwarranted simplicity to the matter strips away layers of depth, rather than enhancing insight.

14,452

It seems to me that one of the most distinctive human characteristics is that of flexibility. 'Single-mindedness is all very well in cows or baboons', wrote Aldous Huxley: 'in an animal claiming to belong to the same species as Shakespeare it is simply disgraceful'.

14,462

'Too much consistency is seeing situations as similar when they are, in fact, different', as John Kay puts it, and as I have tried to suggest.68 Or, to quote Aldous Huxley again: Too much consistency is as bad for the mind as it is for the body. Consistency is contrary to nature , contrary to life. The only completely consistent people are the dead.

14,491

The sharp-cut scientific classifications are essential for scientific method, but they are dangerous for philosophy. Such classification hides the truth that the different modes of natural existence shade off into each other.73

14,765

Moral evaluation in particular involves, or should involve, moving on from what the left hemisphere sees as a simple causal sequence, step by step, to what the right hemisphere sees as a whole narrative: how the person arrived at those particular circumstances, what was going on in his or her mind at the time, what sort of a person would think or act in that way, whether we would like to live in a world where such people form the majority – in other words, the whole human picture, not the mechanistic part picture:

14,807

There are rational goals that the rational person ought to pursue, which nonetheless can only be achieved by not pursuing them: along with sleep and happiness, these include things as diverse as wisdom, sexual performance, sympathy and being natural. The direct approach destroys its object.

14,835

Good decision-makers are eclectic and not wedded to 'consistency'. But when things turn out, for whatever reasons, the way we thought they might, we tend to overestimate our role in their doing so. We overestimate our ability to predict and to control. That way disaster often ensues .

14,859

many rational and desirable goals are simply incompatible with the state of mind required to pursue them: they must come, if they come at all, as the by-products of a life well lived. Among these are humility, courage, love, admiration, faith and understanding. As the philosopher Jon Elster points out, in his brilliant book Sour Grapes (subtitled Studies in the Subversion of Rationality), trying to bring about such states directly is a moral fallacy. The corresponding intellectual fallacy is the attribution of such states, when they occur, to intentional action.

15,015

We can kill creativity by instituting regular ‘procedures’ and closely monitoring the process – all of which the left hemisphere does best. This succeeds in over-controlling the ‘product’, and ensuring mediocrity (though without, of course, avoiding mistakes). It rules out anything not explicit, and already foreseen in the schema. And it discourages analogical thinking: seeing the patterns that come to us through intuition. It’s not new procedures we need: it’s new shapes, new Gestalten – new wholes.

15,246

In the past I was afraid of stepping out of line too much and making a fool of myself. Now that I know how foolish we all are, I am happy to be just another human being.

15,706

The idea of reason has at least two main meanings: a rigid, abstract, linear mode, prizing precision and certainty, and owed to the left hemisphere; and an embodied ability to hold in suspension knowledge of various kinds justly, and to evaluate them in the context of lived experience, where reason is not in any sense at war with feeling or imagination. This mode is more dependent on the right hemisphere: and is, like the right hemisphere’s understanding of science, in my view more fruitful than the alternative that is often served up in its place. And it is vanishing and dying.

15,728

to accept that any statement and its opposite must be true. Bohr makes an important distinction: everyday truths do not give rise to paradox, deep truths do.

16,816

Part of the power of the unconscious mind is its capacity to deal with numerous different elements simultaneously – and, even more importantly, to see their interconnexions one with another. This form of parallel processing is central to pattern recognition because it is able, without obtruding into conscious awareness, to take note of complex simultaneous events, and the emergent patterns of connexion: a seamless process.

16,879

Much of what we call intuition is the result of what John Kay calls 'finely honed, well-developed skills'.85 'Well-developed' here means repeatedly exercised through practice, and it is on the grindstone of experience that our intuitions are honed.

17,167

Intuition appears to be something that, while inevitably fallible, is often more reliable, much quicker, and capable of taking into account many more factors, than explicit reasoning, including factors of which we may not even be consciously aware. It also underlies motor, cognitive and social skills, and is the ground of the excellence of the expert. The attempt to replace it with rules and procedures is a typical left hemisphere response to something it does not understand – a response that is, alas, powerfully destructive. We inhabit a world in which reason is needed more than ever before, yet in which reason is so narrowly conceived that it drives out true understanding. For that we would have had to learn respect for the power of intuition, not as opposed to reason, but as both grounding it, and the means for it to fulfil its potential in making judgments in life.

17,253

First, that reasoning can of course be helpful, but only if you can be certain your reasoning is right – which apparently even mathematical geniuses can't. Failing that, reasoning will mislead you: and being attuned to what experience tells you (pigeon fashion), without letting the attempt to reason get in the way, is safer – though it may take a while longer. Thinking slow?

18,097

The historian Jerry Muller, in his devastating critique of contemporary managerial culture, in academia, in medicine and the other professions, The Tyranny of Metrics, makes many important points. One is the way in which making things explicitly measurable not only, as is very well known, distorts the practice that is being measured, focussing attention, both for the professional and for the bureaucrat, on what is measurable at the expense of the important elements that are not, but how it narrows our sense of what we are dealing with – confirming the point that an intuitive understanding takes into account far more, not less, than the management measures. 'Quantification', writes Muller, 'is seductive, because it organises and simplifies knowledge. It offers numerical information that allows for easy comparison among people and institutions. But that simplification may lead to distortion, since making things comparable often means that they are stripped of their context, history, and meaning.'

18,107

When organisations committed to metrics wake up to this fact, they typically add more performance measures – which creates a cascade of data, data that becomes ever less useful, while gathering it sucks more and more time and resources … Because belief in its efficacy seems to outlast evidence that it frequently doesn't work, metric fixation has elements of a cult. Studies that demonstrate its lack of effectiveness are either ignored, or met with the assertion that what is needed is more data and better measurement. Metric fixation, which aspires to imitate science, too often resembles faith.'

18,114

In situations where there are no real feasible solutions to a problem, the gathering and publication of performance data serves as a form of virtue signalling. There is no real progress to show, but the effort demonstrated in gathering and publicising the data satisfies a sense of moral earnestness. In lieu of real progress, the progress of measurement becomes a simulacrum of success

18,124

'The demand for measured accountability and transparency waxes as trust wanes', writes Muller. 'There is an elective affinity between a democratic society with substantial social mobility and greater ethnic heterogeneity, and the culture of measured accountability … The quest for numerical metrics of accountability is particularly attractive in cultures marked by low social trust.'

18,134

An obsession with counting, and accountability leads, here as elsewhere, to mediocrity.

18,251

Reason's progeny are abstraction, precision and linearity; intuition's progenitors are embodiment, the intrinsically imprecise, and complexity. Intuitions are of different kinds, but they are all subject to being discounted by a culture that relies overmuch on the left hemisphere's take. Once again, the more reliable of them appear to depend on right hemisphere function.

18,257

In our culture, all mores have been abandoned; and what should remain implicit and in the realm of embodied skill is foregrounded as a 'problem' to be consciously solved – with the result that we grossly simplify and omit what is beyond calculation.